DSA03-Natural Language Processing

List of Lab Program

3. Write program demonstrates how to perform morphological analysis using the NLTK library in Python.

Code:

```
import nltk
from nltk.stem import PorterStemmer
from nltk.stem import WordNetLemmatizer
nltk.download('wordnet', quiet=True)
nltk.download('omw-1.4', quiet=True)
words = ["playing", "played", "studies", "cars"]
stemmer = PorterStemmer()
lemmatizer = WordNetLemmatizer()
print(f"{'Original Word':<18}{'Stem':<12}{'Lemma':<12}")
print("-" * 42)
for word in words:
    stem = stemmer.stem(word)
    lemma = lemmatizer.lemmatize(word)
    print(f"{word:<18}{stem:<12}{lemma:<12}")
```

Output:

```
Original Word     Stem        Lemma
------------------------------------------
playing           play        playing
played            play        played
studies           studi       study
cars              car         car
```